Ashwani Kumar Dwivedi

Experienced Python & Django Developer with overall 2 years of expertise in developing scalable web applications, RESTful APIs, and e-commerce platforms. Skilled in Django, DRF,, Docker, and AWS (EC2, S3, RDS) for deploying robust, production-ready systems. Well-versed in data processing with Pandas and NumPy, and building cloud-native applications. Recently focused on Generative AI (GenAI), developing intelligent agents using LangChain, LLMs, and RAG (Retrieval-Augmented Generation) for advanced chatbot and automation use cases.

Backend Developer

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EDUCATION

Master Of Computer Application
LNCT BHOPAL

07/2021 - 05/2023 *7 .95 CGPA*

SKILLS

Python | Django | MYSQL | DRF | Pandas | NumPy | AWS | GIT | GITHUB | GENAI | AgenticAI | Langchain | LLM's | VectorDB

WORK EXPERIENCE

Software Developer-Associate
SmartSchool Education Private Limited

12/2023 - Present *Noida,India*

SmartSchool – Feature-Rich E-commerce & Learning Platform ***Role :*** *Full-Stack Developer* ***Tech Stack****: Python, Django, Django REST Framework (DRF), Bootstrap, jQuery, Celery, Redis, Docker, Docker Compose, AWS (EC2, S3, RDS, CloudFront, CloudWatch), PostgreSQL/MySQL, Django Channels, WebSockets, Pandas, NumPy, LLM, LangChain, RAG, Git, CI/CD (GitHub Actions), Stripe/Razorpay/PayPal* ***Key Contributions****: Developed and deployed* ***SmartSchool****, a robust* ***e-commerce and learning platform*** *using* ***Python****,* ***Django****, and* ***DRF****, featuring scalable APIs for product management, order workflows, and* ***real-time delivery tracking****. Implemented* ***pagination****,* ***filtering****,* ***query optimization****, and* ***role-based access control (RBAC)*** *using* ***DRF permissions*** *for secure and scalable operations. Enhanced* ***Django Admin*** *for operational efficiency and integrated* ***Razorpay, Stripe, and PayPal*** *for seamless, secure payment transactions. Built a* ***responsive frontend*** *using* ***Django Templates****,* ***Bootstrap****, and* ***jQuery****, ensuring mobile-first UI/UX. Integrated* ***Celery + Redis*** *to handle background tasks (order confirmation emails, inventory sync, analytics). Enabled* ***real-time updates*** *(order status, admin notifications) using* ***Django Channels + WebSockets****. Leveraged* ***Pandas*** *and* ***NumPy*** *for generating user behavior and sales analytics dashboards. Used* ***LLMs and LangChain agents*** *to build an internal AI assistant for handling support queries and summarizing user behavior using* ***RAG*** *pipelines. Ensured* ***high availability*** *and* ***zero-downtime deployments*** *via* ***Django migrations****,* ***Docker containers****, and* ***Docker Compose*** *orchestration. Deployed on* ***AWS cloud*** *using:* ***EC2*** *for backend hosting* ***S3*** *for static/media storage* ***RDS (MySQL)*** *for scalable DB* ***CloudFront*** *for fast content delivery* ***CloudWatch*** *for monitoring & logging Built and automated* ***CI/CD pipelines*** *using* ***GitHub Actions****, streamlining deployment and updates. Applied* ***advanced indexing****, database tuning, and* ***AWS backup strategies*** *to ensure platform performance and reliability.*

Achievements/Tasks

- Backend- **Django,DRF,ORM,JWT AUTH,Pagination**
- Database-**MYSQL**
- API Testing - **Postman**
- Version Control - **GIT , GITHUB**
- Cloud - **AWS(IAM,EC2,S3,RDS)**
- Hosting Platform - **AWS**
- Extra - **Docker,Linux,Numpy,Pandas,LLM,Langchain,GENAI**

PERSONAL PROJECTS

RAG Based Chatbot Using OpenAI,Langchain,VectorDB (03/2025 - 06/2025)

- **Project Summary**: Designed and implemented a **Retrieval-Augmented Generation (RAG)** based chatbot that provides accurate and context-aware answers by combining **OpenAI LLMs** with a **vector-based document retrieval system**. Leveraged **LangChain** to orchestrate the LLM and vector store integration, enabling users to upload documents (PDF/TXT) and query them conversationally. Used **FAISS/ChromaDB** to store document embeddings for fast semantic search. Employed **LangChain**'s retrieval and chaining features to construct a dynamic question-answering pipeline. Embedded documents using **OpenAI Embeddings** or **HuggingFace Transformers** for retrieval accuracy. Developed a frontend using **Streamlit** (or API with **FastAPI**) for user-friendly interaction. Added **Pandas** and **NumPy** to analyze and summarize tabular content within the documents. Applied **chunking and metadata tagging** to handle large files and maintain retrieval precision. Outcome: Delivered a fully functional GenAI-powered assistant capable of answering domain-specific questions from uploaded documents in real time — ideal for use cases like medical, legal, or academic document Q&A.
- Github

IT HUB (07/2023 - 09/2023)

- **Tech Stack**: Python, Django, Django REST Framework (DRF), MySQL, Bootstrap, HTML/CSS, JavaScript **Project Summary**: Designed and developed a robust **IT Hub Management System** using **Python**, **Django**, and **Django REST Framework**, aimed at streamlining internal operations and data management for tech companies. Built secure and scalable **CRUD APIs** for managing company resources such as employees, projects, assets, and client data. Integrated **MySQL** for structured data storage and optimized complex queries for fast and reliable data retrieval. Ensured **modular design** with clean separation of concerns across models, views, serializers, and permissions. Implemented **role-based access control (RBAC)** using DRF's permission classes to enforce data security. Designed a responsive and user-friendly admin dashboard using **Bootstrap** and **Django templates** for real-time monitoring and updates. Enabled filtered data views, search functionality, and relational data management for seamless navigation and usability. Focused on **performance, security, and maintainability**, ensuring the system can scale with growing business needs. **Outcome**: Delivered a complete IT management solution that significantly improved data accessibility, operational transparency, and digital workflow efficiency across departments.
- **Github**